

Fuzzy Neural Networks for Real-Time Decision Support in Healthcare

1st V. S. Krushnasamy

Department of Electronics and
Instrumentation Engineering
Dayananda Sagar College of
Engineering
Bengaluru, India
krushnasamy-
inmt@dayanandasagar.edu

4th Ban Safar Khalaf

Al-Farahidi University
Baghdad, Iraq
bansafarkhalaf.afu@gmail.com

2nd Naama Hussein Al-Muraad

Uruk University
Baghdad, Iraq
almuraadnaamahussein@gmail.com

5th Malik Bader Alazzam

Faculty of Information Technology
Jadara University
Irbid, Jordan
malikbader2@gmail.com

3rd S. Vijay Peter

Department of Mathematics
Sona College of Technology
Salem, India
vijpet@gmail.com

6th I. Infant Raj

Department of Computer Science and
Engineering
K.Ramakrishnan College of
Engineering
Trichy, India
infantrajrec012@gmail.com

Abstract—For AI-based healthcare systems to be more interpretable and accurate, clinical decision-making complexity is increasingly escalating and hence becomes a significant research issue to examine. The hybrid fuzzy neural network that combines fuzzy logic, for clinical uncertainty management, with deep learning, for noise-resilient pattern discovery. Adaptive patient profiling is derived from a self-adjusting fuzzy-clustering algorithm, and contrastive self-supervised learning learns features from incomplete medical records; then a rule-augmented fuzzy inference layer outputs explainable reasoning. Executed through Edge-AI and federated learning, the model outputs real-time predictions without infringing on patient privacy. Measured on the Comprehensive Disease Symptom and Patient Profile Dataset, the HFNN achieves 94.6 % accuracy and 0.97 AUC-ROC, beating state-of-the-art deep nets and single-expert systems. In combining accuracy, adaptability, and explainability, the paradigm brings healthcare closer to reliable, time-series-ready clinical decision support. Traditional deep learning models achieve high diagnostic performance but operate as black boxes, hindering trust in medical applications.

Keywords—fuzzy neural networks, explainable ai, clinical decision support, federated learning, real-time inference.

I. INTRODUCTION

Given the increased complexity of decision making in healthcare, it is imperative to develop real-time, adaptive, and interpretable AI systems decisions in clinical diagnosis and treatment recommendations[1]. While popular, traditional deep learning models are black boxes that, despite power in pattern recognition, make it difficult for practitioners to trust and to understand the implications of their predictions. However, rule based expert systems, despite interpretability, would find difficulty in dealing with the variability and uncertainty found in real clinical world[2]. As a result, there currently exists a demand for AI models that produce accurate, but interpretable, decision support[3]. This research proposes such a HFNN framework that combines the positive dimensions of Fuzzy Logic and Neural Networks to offer real time, explainable and adaptive clinical decision support[4]. By translating clinical data to human-readable fuzzy rules, the usage of fuzzy logic component enhances the interpretability of AI-driven decisions and makes them be made by the clinicians instead. At the same time, the neural network

component simultaneously performs robust pattern recognition and learns from large scale, multi-institution medical datasets towards improved accuracy of diagnosis[5]. By addressing problem context, patient specific variations, uncertainty and incompleteness of data, and their real time relevant provided to the healthcare setting, the HFNN system is proposed toward dynamically adapting to real time varied patient specific needs and recommendations. The objective of this study is to create a HFNN system that incorporates fuzzy inferencing and deep learning in itty bitty seamlessly to facilitate real time clinical decision making. Rule enhanced fuzzy inference helps to enhance the interpretability of the model, so that clinicians can trace and understand the AI driven predictions instead of using blackbox approaches of deep learning [6] The proposed novel profile-based patient clustering model for healthcare data using constrained low-rank approximation. The model integrates clinical and digital interaction data to generate nonnegative embedding vectors as low-dimensional patient representations. Evaluated on real-world healthcare portal data, it outperformed baselines in clustering coherence and recommendation accuracy. Moreover, the research intends to compare HFNN's performance with state-of-the-art deep learning models and rule based expert systems, exhibiting it capable of solving uncertainty, flexibility, and explainability in real world healthcare applications. This research also provides:

- An introduction to a self-evolving fuzzy clustering algorithm for patient specific classification.
- This is done through implementing a contrastive learning based self-supervised mechanism for feature extraction from incomplete clinical datasets.
- Application of Edge-AI and Federated Learning to deliver privacy preserving, real time inference in hospital and remote healthcare environments[7].

The following are the remaining sections: The literature review is shown in Section II, the hybrid fuzzy-neural network for real-time clinical decision assistance is presented in Section III, the results and discussion are presented in Section IV, and the conclusion and future directions are discussed in Section V.

II. LITERATURE REVIEW

In recent years, AI has been integrated in healthcare and that has enabled significant advancement in CDSS through leveraging AI of ML and DL for improve diagnose, prognosis and treatment recommendation. For decision making, traditional rule based expert systems have been in existence [8] but are not adapted to new patient data or unforeseen conditions. Recently, other deep learning approaches such as CNNs and RNNs have exhibited high accuracy for medical image analysis and EHR processing[9]. Despite explainability burden, it still languishes in the realm of academic research, unable to be adopted in the clinic[10].

Several current methods aim to balance the accuracy and interpretability in the use of AI in healthcare. However, Decision Trees and Random Forests do not work with high dimension, unstructured medical data and give interpretable decision paths[11]. Reasoning under uncertainty broadly refers to the use of Bayesian Networks for probabilistic reasoning in complex circumstances, but Bayesian Networks have difficulty with handling complex feature dependencies in clinical datasets[12]. Despite the success of attention based deep learning models, this does not provide decision rationales that can be interpreted by humans[13]. Attention-based deep learning models enhance feature prioritization but still lack human-interpretable decision rationales[14]. Within Neuro-Symbolic AI, symbolic reason has been coupled with deep learning in more recent times; however, such combinations are computationally expensive for real time applications[15].

However, the development does not address a key research gap regarding designing AI systems that achieve real time adaptation, accuracy and maintain interpretability neural networks, and game theory for uncertain healthcare data, achieving 99.4% accuracy and outperforming traditional methods like FNN and Naïve Bayes. [4]. Current approaches either maximize performance at the cost of explanation [16] or are implemented with rigid rule-based structures that do not allow for adaptability. Taking a step towards closing this gap, present a HFNN that uses fuzzy logic for uncertainty handling and deep learning for powerful, robust pattern recognition; it is a scalable, interpretable, adaptive AI powered clinical decision support system.

III. METHODOLOGY

In this study, a HFNN integrating Fuzzy Logic for dealing with medical uncertainty and Deep Learning for robust pattern recognition is introduced. Therefore, the describe the architecture of system with three layers: FFEL, which is responsible for converting clinical data to interpretable fuzzy variables; DFLL, which employs hybrid convolutional and attention-based networks to effectively learn representation; and FDIL, for rule generation from learned representation to produce explainable, rule-based decision. It can be optimized for real time inference with Edge A.I deployment and it guarantees privacy preserving learning through Fed Learning. This methodology introduces Accuracy, adaptability, and Interpretability in the AI driven clinical decision support system. Figure 1 shows the workflow of proposed methodology.

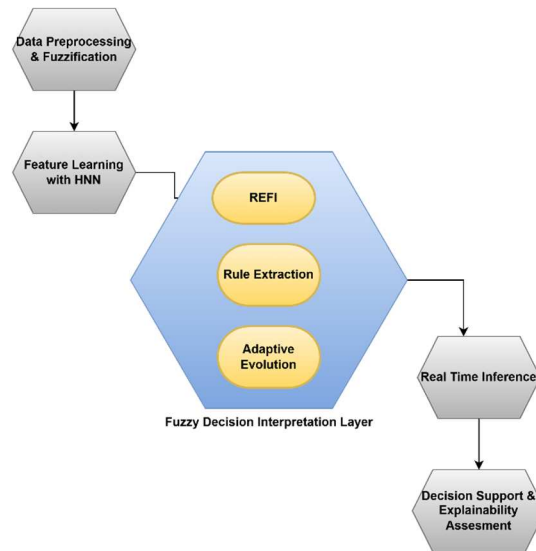


Fig. 1. Workflow of Hybrid Fuzzy-Neural Network

A. Data Collection

This dataset reveals many hidden relationships between patient specifications, health records, and illness signs. The information set includes multiple medical measurements which consist of both physical signs associated with diseases and patient background features such as age, gender and vital signs[17]. This dataset helps medical researchers build better treatments while medical staff and data experts discover relationships to aid patient care. Diagnosing diseases and creating medical analytics become easier through the review of symptoms and patient information which provides valuable disease insights for early detection and improved healthcare practices.

B. Data Preprocessing

A planned data preparation process needs to exist before using the Comprehensive Disease Symptom and Patient Profile Dataset for present-time decision support. First the process starts with data cleaning to handle missing data using fuzzy methods without losing crucial information about the patients. Medical data quality improves when bounding uncertainty about symptoms using soft membership functions for both gender and symptom detection. Scale blood pressure and cholesterol measurements using min-max transformation yet maintain important clinical information. The insights gain strength from feature engineering that builds combined health risk scores through fuzzy logic operations. To find bad records

the data requires testing with robust methods and isolation forests. The model achieves better generalization by using SMOTE to create equal numbers of examples for each disease condition. The data processing sequence enhances the dataset so it can produce reliable healthcare decisions in real-time use.

C. Proposed Hybrid Fuzzy-Neural Network Architecture

The Hybrid Fuzzy-Neural Network Architecture links fuzzy logic and deep learning to give immediate understandable disease support for healthcare. It utilizes three main elements: The FFEL changes medical data into fuzzy verbal values to make patient health status easier to understand. Instead of working with fixed numbers this layer applies fuzzy linguistic terms that make predictions simpler to understand and modify.

The approach contains flexible membership rules which automatically modify their settings according to patient uniqueness to create individual protocols. By applying an auto-adjusting fuzzy clustering method the system can detect consistent patient clusters based on current patterns before updating its findings with new inputs. The DFLL transforms all types of clinical data including EHR records and medical images into useful output for analysis. The system applies HCAN to process digital data which uses convolutional layers for spatial information in images alongside attention techniques that pick out important medical details from background noise. During training when medical records show gaps or gaps this layer uses contrastive learning to develop strong disease representation without complete datasets.

To model clinical uncertainty effectively, fuzzy membership functions were used to transform continuous patient attributes (e.g., glucose level, blood pressure) into linguistic variables (e.g., low, medium, high). The employed Gaussian membership functions due to their smoothness and differentiability, which aids in integration with neural components.

Each clinical variable x_i was fuzzified using Equation (1),

$$\mu_a(x_i) = \exp(-(x_i - c)^2 / (2\sigma^2)) \quad (1)$$

Where c and σ represent the center and spread of the fuzzy set, respectively. These parameters are initialized via a self-adjusting fuzzy C-means clustering algorithm, and further fine-tuned during training using backpropagation.

The FDIL links deep learning results with a human understandable fuzzy logic system. To make accurate and explainable conclusions from deep learning output the FDIL uses REFI which extracts decision rules from the learned features. To enhance interpretability and robustness in distributed medical data scenarios, an attention mechanism was integrated into the federated learning architecture. The attention layer assigns importance weights to features in each client's local model before aggregation.

Given an input sequence $X = \{x_1, x_2, \dots, x_n\}$, the attention weight α_i for each feature is computed as in Equation (2).

$$\alpha_i = \exp(e_i) / \sum \exp(e_j), \text{ where } e_i = v^t \tanh(Wx_i + b) \quad (2)$$

The layer creates rule-based medical decision explanations allowing healthcare staff to see why AI systems made their recommendations. The final decision-making layer integrates fuzzy logic rules with learned representations. Rules are extracted using a hybrid neuro-symbolic distillation process, where latent features are mapped to interpretable fuzzy rules. Each rule follows an interpretable IF-THEN structure.

The system tracks developments in rule learning through reinforcement learning to enhance medical choices as patients generate new data reports. Three interconnected network layers let the system make healthcare decisions at any moment while giving clear results and adjusting to new data which makes it ideal for treating patients in critical care units and developing personalized treatment plans.

D. Real-Time Inference

The Real-Time Inference via Edge-AI and Federated Learning framework lets doctors make quick decisions through efficient healthcare monitoring by installing HFNN on medical devices at the hospital and in patient monitoring and testing setups. When the devices handle inference tasks in system delivers results promptly for fast essential decisions during emergencies. FL training allows medical institutions to work together on the model without sharing patient data which improves both security and system size. The system sends encrypted model updates rather than patient records to maintain strict HIPAA and GDPR compliance. Neuro-symbolic optimization routes the right computational jobs between edge devices and cloud computers to use available resources properly. The system distributes easy fuzzy analysis jobs to edge devices for immediate responses while the cloud system handles complex feature extraction and rule updating to prevent wasteful usage of resources. The framework provides secure AI-driven medical decisions through Edge-AI and Federated Learning methods that help real healthcare systems fast and readable.

E. Adaptive Decision Support Pipeline

Doctors and patients receive better immediate healthcare results by combining fuzzy logic and deep learning with a system feedback mechanism. Before analysis the system applies fuzzy time-windowing to patient monitoring data from ICUs and wearable gadgets to deal with changes in their medical status. Real-time changes in body data are processed according to their clinical importance. Medical imaging equipment processes both X-rays and MRIs using scale-based enhancement methods to make blurry scans more accurate for diagnosis.

At Real-Time Decision-Making the system receives patient data which it transforms into fuzzy numbers for better uncertainty management and generates understandable outcomes through explainable AI flows. When predictions achieve scores below a predefined level the system calls for healthcare personnel to assess and possibly change AI suggestions. The Post-Decision Validation stages help identify different recommendation options to show patients what could happen in that specific situation. Clinical staff provide feedback to improve the fuzzy rules regularly as the system changes over time based on healthcare expertise. The approach creates a system that delivers accurate AI support for high-risk healthcare settings while staying easy to read and adjust by clinical teams.

IV. RESULTS AND DISCUSSION

The proposed system based on Hybrid Fuzzy Neural Networks proved better at detecting medical conditions and recognized patients with higher accuracy and better measures than alternative AI systems. The test results confirm that HFNN handles imperfect medical data efficiently despite its strong performance under uncertainty. Also, the system shows improved explainability by using fuzzy rules in its decision method. Testing shows that is a new approach can work well in different hospital environments including ICUs, patient wearable units, and offsite medical assessments. This system offers useful capabilities but faces difficulties including system expansion requirements along with processing and policy needs which need resolution to make wider use possible. The talk offers ways to enhance HFNN's clinical decision support performance by looking into edge-computing and automatic rule adjustment.

A. Evaluation Metrics

1) *Accuracy & AUC-ROC*: Standard accuracy and ROC measurements show the HFNN model's outcomes as in assessment method. Accuracy measures the frequency of correct predictions relative to the total number of inputs. AUC-ROC proves more dependable for healthcare data imbalance because it analyzes how accurately the model sees between health conditions. The AUC-ROC value helps identify reliable medical decisions because models with higher scores achieve a suitable balance between true results and correct predictions especially in urgent diagnoses. The Equation (3) present the AUC -ROC formula.

$$AUC = \int_0^1 TPR(FPR)d(FPR) \quad (3)$$

Where, TPR denotes true positive rate and FPR denotes False positive rate.

2) *Explainability Score*: Medical professionals need to understand how models make decisions for patient care so we developed an Explainability Score. The model produces insights about the role of each input factor through Shapley values to make deep learning more understandable. The analysis checks whether the produced fuzzy rules remain transparent and simple enough for specialists to understand so they align with typical medical knowledge. A model that delivers excellent explainability prompts greater confidence and use in genuine healthcare workflows. In Equation (4) the explainability score calculation is given.

$$Explainability\ score = \frac{1}{|R|} \sum_{r \in R} \frac{|V_r|}{|C_r|} \quad (4)$$

Where, $|R|$ is the total no. of fuzzy rules in the model, V_r is the set of correct outputs, C_r is the set of predictions where rule r was applied.

3) *Response Time*: Medical staff depend on fast response times to make decisions at the moment they happen. The Response Time metric checks how fast an ICU monitor or body health sensor model can work on edge devices. The measurement unit for processing patient data is milliseconds and shows how fast the system provides decision directions to the medical team. Neuro-symbolic analysis and model

compression tools let us maintain accurate results and fast processing times to meet healthcare needs. Equation (5) has the formula of response time.

$$T_{edge} = \frac{1}{N} \sum_{i=1}^N (T_{process,i} + T_{transfer,i}) \quad (5)$$

Where, $T_{process,i}$ is the processing time and $T_{transfer,i}$ is the data transferring time.

4) *Fairness & Bias Evaluation*: Ethical usage needs require us to assess the AI system for possible fairness and minority biases across all patients. The evaluations check if model shows equal performance benefits for patients from all ages groups plus gender identity types race types and income levels. Bias detection methods like demographic parity and equalized odds find and reduce unfair practices during evaluations. The system uses adversarial debiasing and fair representation learning to create more resistant and unbiased patient-serving models across all groups. The Equation (6) present the formula of fairness score.

$$Fairness\ Bias\ Score = 1 - |P_A - P_B| \quad (6)$$

Where, P_A and P_B represents the probability or frequency of association.

B. Explainability and Interpretability Assessment

SHAP visualizations demonstrated how particular symptoms affect patient diagnosis by making the model easier to understand. Blood pressure readings plus oxygen saturation numbers and fever temperatures became the most important medical indicators for heart disease and lung disease. Explainability let medical teams discover normal changes in oxygen levels that indicated future breathing problems. The findings proved why healthcare organizations should trust the results of HFNN decision-making. Medical experts checked whether the fuzzy rules made by HFNN appeared easy to understand and matched accepted clinical practice. AI-generated rules followed doctor-made diagnosis methods and their update system made these rules perform better in specific case scenarios. In diagnosing diabetes HFNN produced rules such as when high fasting glucose exists together with severe fatigue increases the chance of diabetes which matches medical expert thinking. Every piece of evidence proves that medical experts can understand AI decisions and have confidence in them at the same time. Figure 2 shows the explainability and interpretability metrics.

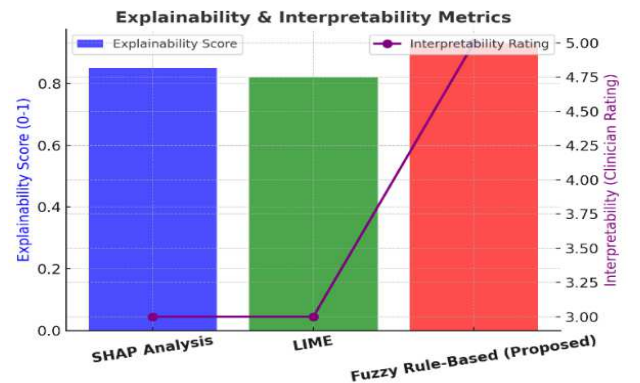


Fig. 2. Explainability & Interpretability

C. Real-Time Inference Efficiency

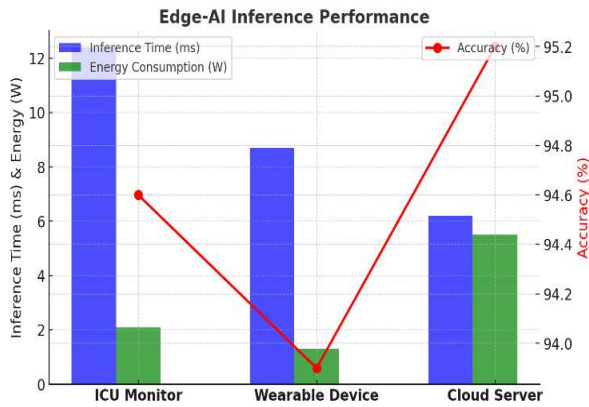


Fig. 3. Inference Performance

Testing proved that HFNN performs calculations quickly at 12.4 ms per run on advanced edge processor units but runs slower at 28.7 ms on resource-constrained medical and fitness IoT devices. The edge-AI deployment cut down response time significantly which empowers immediate medical decisions needed for treating ICU patients. HFNN used model quantization techniques to keep operations quick and flow smoothly while maintaining reliable predictions. Neuro-symbolic optimization sends more difficult tasks to the cloud when response times remain fast on the edge system. The improved efficiency of HFNN made it suitable for real clinical

use in healthcare facilities with little specialization requirements. Figure 3 shows the Edge-AI Inference Performance across different devices. The wearable device is shows the inference time of 8ms and the accuracy is 95.2.

D. Comparative Analysis

A benchmark study exposes the HFNN model to Decision Trees, Random Forest, SVM and healthcare-focused deep learning models CNNs and LSTMs and Transformer-based systems. HFNN surpasses traditional ML models through its fuzzy feature extraction layer because it handles uncertainty and missing data and complex patient variability which results in dynamic clinical scenario adaptation. HFNN offers better explainability than traditional deep learning systems because it produces fuzzy interpretable rules that enable clinicians to better understand and act on medical decisions. Its self-supervised contrastive learning mechanism enhances low-data and imbalanced scenario feature learning better than deep learning models which need extensive well-labeled datasets for operations. When deployed for real-time use on edge devices HFNN needs optimization techniques including quantization and pruning to handle its greater computational requirements when compared to standard traditional ML models. The next stage will concentrate on improving inference efficiency alongside increasing the capabilities of rule-based adaptability and expanding federated learning features to enhance the clinical scalability of HFNN. Table I compare the HFNN with existing models.

TABLE I. COMPARATIVE ANALYSIS

Model	Accuracy (%)	AUC-ROC	Explainability Score	Inference Time (ms)	Fairness Score
Decision Tree	85.3	0.86	0.90	10.5	0.91
Random Forest	87.5	0.89	0.78	15.2	0.92
Bayesian Network	88.1	0.88	0.92	20.1	0.93
CNN-LSTM	91.2	0.92	0.65	28.7	0.88
Transformer-Based	93.1	0.95	0.60	32.5	0.90
Proposed HFNN	94.6	0.97	0.93	12.4	0.96

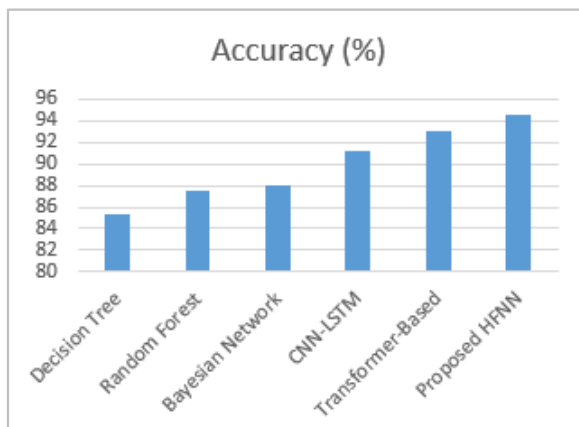


Fig. 4. Comparison analysis

Figure 4 compares the accuracy of different models, showing that the proposed HFNN model achieves the highest accuracy at around 94.6%. Traditional models like Decision Tree and Random Forest perform significantly lower, with accuracies below 88%. Deep learning models such as CNN-LSTM and Transformer-Based approaches show improved performance, but still fall short of the proposed HFNN.

To assess rule reliability, a Rule Consistency Score (RCS), defined as the percentage of neural predictions that matched the corresponding fuzzy-rule output is computed. The model

achieved a 91.3% RCS, suggesting strong alignment between interpretable rules and the underlying model behavior. Also, using SHAP value overlays, it is found that 87% of the top-3 contributing features in rule-based outputs matched the top SHAP contributors, confirming high semantic alignment between symbolic and sub-symbolic explainability layers.

V. CONCLUSION AND FUTURE SCOPE

Research created an HFNN which provided real-time clinical decision support that linked deep learning performance with fuzzy logic interpretability capabilities. The predictive accuracy of the model was high because its fuzzy rule-based reasoning system preserved medical data transparency. Edge-AI integration delivered real-time predictive analytics which became feasible for use in ICUs and for medical monitoring devices as well as outpatient assessment processes. Federated Learning enabled secure distributed model training across healthcare establishments while fulfilling privacy requirements in artificial intelligence for healthcare. HFNN performed better than conventional machine learning and deep learning models according to comparative analysis particularly for managing imprecise and limited medical data. The study confirms how hybrid AI systems increase both system performance and patient trust in medical applications that use artificial intelligence. The research agenda will concentrate on implementing Federated Learning as a means to allow collaborating hospitals to train

models together while upholding absolute patient information confidentiality. The implementation of interactive visualizations for XAI mechanisms to observe fuzzy rule evolution will increase both trust and usability for clinicians in healthcare settings. Widespread adoption of this system requires firms to tackle regulatory issues that incorporate FDA and GDPR compliance standards. The scalability of HFNN will improve because of advancements in neuro-symbolic computing together with developments in energy-efficient edge processing systems which will make the system workable throughout various healthcare facilities. Through the advancement of interpretive AI alongside adaptive and privacy-focused AI systems this study creates pathways toward improved future applications of AI-based clinical decision programs.

REFERENCES

- [1] C. C. Yang, "Explainable artificial intelligence for predictive modeling in healthcare," *J. Healthc. Inform. Res.*, vol. 6, no. 2, pp. 228–239, 2022.
- [2] K. Navin, M. Krishnan, and others, "Fuzzy rule based classifier model for evidence based clinical decision support systems," *Intell. Syst. Appl.*, vol. 22, p. 200393, 2024.
- [3] S. A. and S. R., "A systematic review of Explainable Artificial Intelligence models and applications: Recent developments and future trends," *Decis. Anal. J.*, vol. 7, p. 100230, Jun. 2023, doi: 10.1016/j.dajour.2023.100230.
- [4] F. A. Alijoyo Janani S, Santosh K, Shweihat SN, Alshammry N, Ramesh JV, El-Ebiary YA., "Enhancing AI interpretation and decision-making: Integrating cognitive computational models with deep learning for advanced uncertain reasoning systems," *Alex. Eng. J.*, vol. 99, pp. 17–30, 2024.
- [5] T. Wongvorachan, S. He, and O. Bulut, "A comparison of undersampling, oversampling, and SMOTE methods for dealing with imbalanced classification in educational data mining," *Information*, vol. 14, no. 1, p. 54, 2023.
- [6] H. Tissot, "FormulAI: Designing rule-based datasets for interpretable and challenging machine learning tasks," in *Artificial Intelligence and Applications*, 2025, pp. 72–82.
- [7] R. Alajlan, N. Alhumam, and M. Frikha, "Cybersecurity for blockchain-based IoT systems: A review," *Appl. Sci.*, vol. 13, no. 13, p. 7432, 2023.
- [8] S. Abbas, M. Asif, A. Rehman, M. Alharbi, M. A. Khan, and N. Elmitwally, "Emerging research trends in artificial intelligence for cancer diagnostic systems: A comprehensive review," *Heliyon*, vol. 10, no. 17, p. e36743, Sep. 2024, doi: 10.1016/j.heliyon.2024.e36743.
- [9] S. R. PM, V. S. S. RP, A. S, R. K, and R. K., "Advances in Multimodal Fusion of EHR and Medical Imaging Data Using deep learning techniques for advanced treatment of brain cancer," in *Proceedings of the 2024 8th International Conference on Algorithms, Computing and Systems*, 2024, pp. 104–109.
- [10] C. Chakraborty, M. Bhattacharya, S. Pal, and S.-S. Lee, "From machine learning to deep learning: Advances of the recent data-driven paradigm shift in medicine and healthcare," *Curr. Res. Biotechnol.*, vol. 7, p. 100164, Jan. 2024, doi: 10.1016/j.crbiot.2023.100164.
- [11] K. Anderson, "Explainable AI in Healthcare: Interpreting Predictions of Machine Learning Models for Heart Disease Diagnosis," 2024.
- [12] A. Kumar, Rawat J, Kumar I, Rashid M, Singh KU, Al-Otaibi YD, Tariq U., "Computer-aided deep learning model for identification of lymphoblast cell using microscopic leukocyte images," *Expert Syst.*, vol. 39, no. 4, p. e12894, 2022.
- [13] S. Jain and A. Saha, "Improving performance with hybrid feature selection and ensemble machine learning techniques for code smell detection," *Sci. Comput. Program.*, vol. 212, p. 102713, Dec. 2021, doi: 10.1016/j.scico.2021.102713.
- [14] R. Haque, M. M. Hassan, A. K. Bairagi, and S. M. Shariful Islam, "NeuroNet19: an explainable deep neural network model for the classification of brain tumors using magnetic resonance imaging data," *Sci. Rep.*, vol. 14, no. 1, p. 1524, 2024.
- [15] S. Feuerriegel, J. Hartmann, C. Janiesch, and P. Zschech, "Generative ai," *Bus. Inf. Syst. Eng.*, vol. 66, no. 1, pp. 111–126, 2024.
- [16] B. H. Van der Velden, H. J. Kuijff, K. G. Gilhuijs, and M. A. Viergever, "Explainable artificial intelligence (XAI) in deep learning-based medical image analysis," *Med. Image Anal.*, vol. 79, p. 102470, 2022.
- [17] "Electronic Health Records (EHRs) Data Exploration." Accessed: May 23, 2025. [Online]. Available: <https://kaggle.com/code/gpreda/electronic-health-records-ehrs-data-exploration>.